

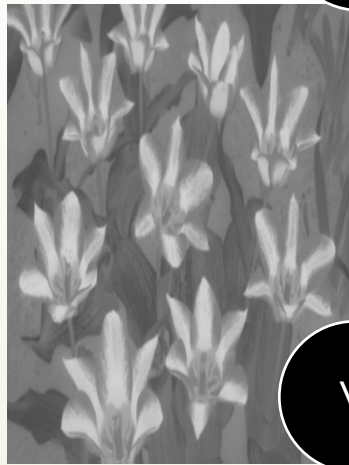
4:2:2 Sub-sampling



RGB



Y

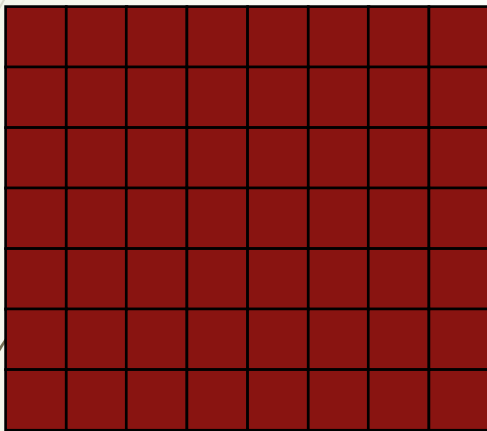


V

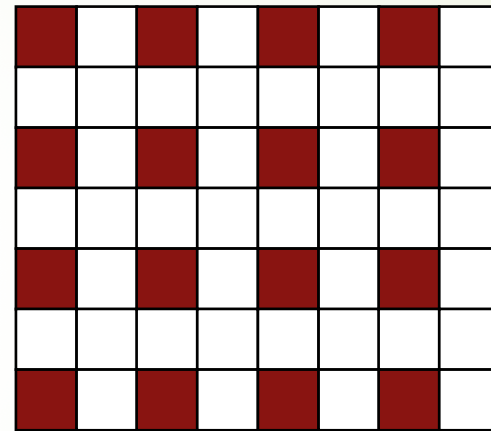


U

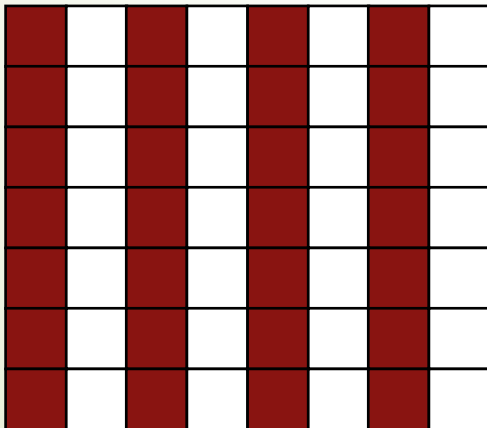
Common Color Subsampling Schemes



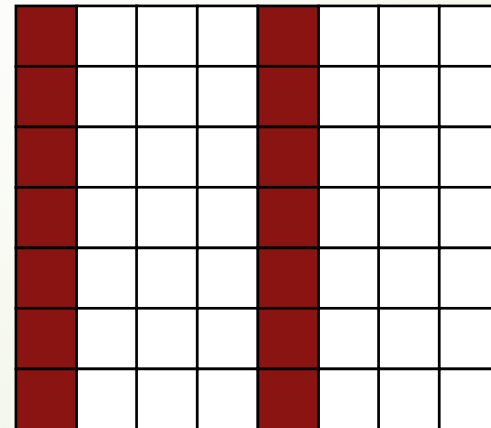
4:4:4



4:2:0



4:2:2



4:1:1

Tools #1: Discrete Cosine Transform

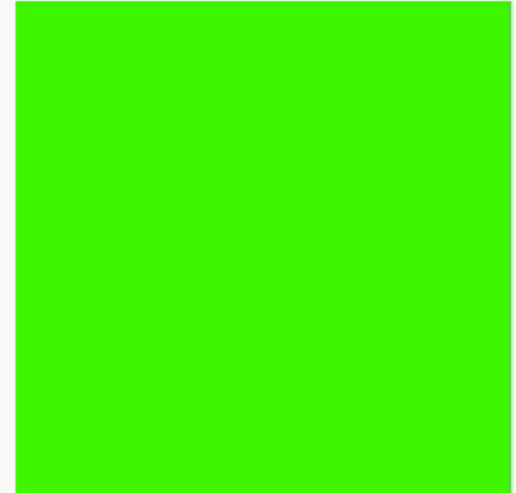
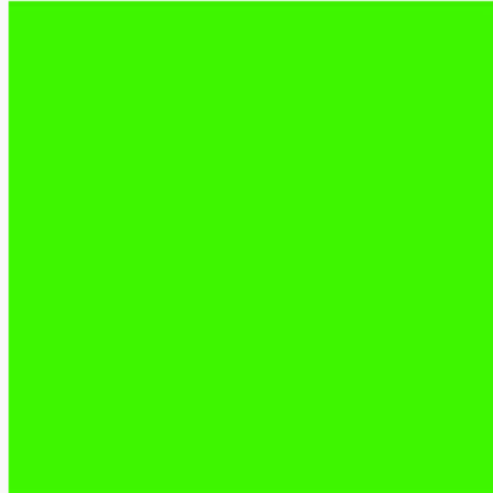
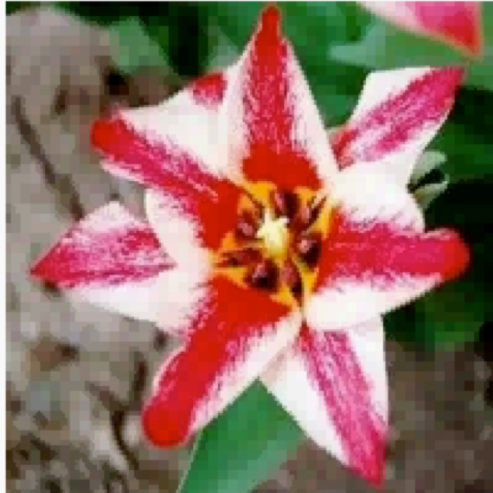


Image Domain

Frequency Domain

Zig-zag Ordering

**So that lower frequency component
are at the upper-left corner**

27	3	1	0
2	1	0	0
4	0	1	0
0	0	0	0

27, 3, 2, 4, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0

Tool #2: Quantization

Q: Why the values of Quantization Table toward lower-right corner are larger?

The diagram illustrates the quantization process using a 4x4 matrix operation:

$$\begin{bmatrix} 242 & 65 & 23 & 5 \\ -54 & -10 & -4 & -2 \\ 13 & 6 & 3 & 5 \\ 2 & 1 & -1 & -2 \end{bmatrix} \div \begin{bmatrix} 8 & 8 & 8 & 8 \\ 8 & 8 & 8 & 16 \\ 8 & 8 & 16 & 32 \\ 8 & 16 & 32 & 64 \end{bmatrix} = \begin{bmatrix} 30 & 8 & 2 & 0 \\ -6 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The resulting matrix is divided into two parts:

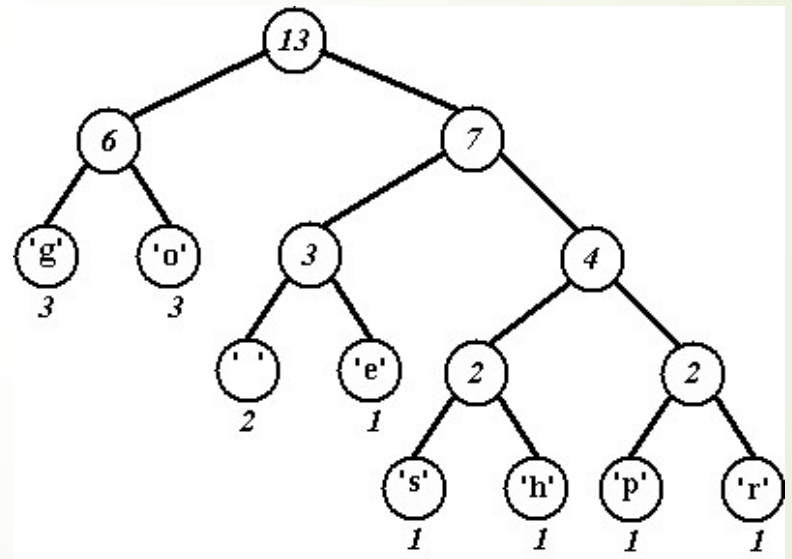
- DC** (Direct Current): The top-left element (30), representing the average value.
- AC** (Alternating Current): The remaining elements, representing the detail or texture of the image.

Quantization Table

Tool #3: Entropy Coding

Run-length coding + Huffman coding

27	3	1	0
2	1	0	0
4	0	1	0
0	0	0	0



27, 3, 2, 4, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0

(27, 1) (3, 1) (2, 1), (4, 1), (1, 2), (0, 5), (1, 1), (0, 4)

Summary of JPEG

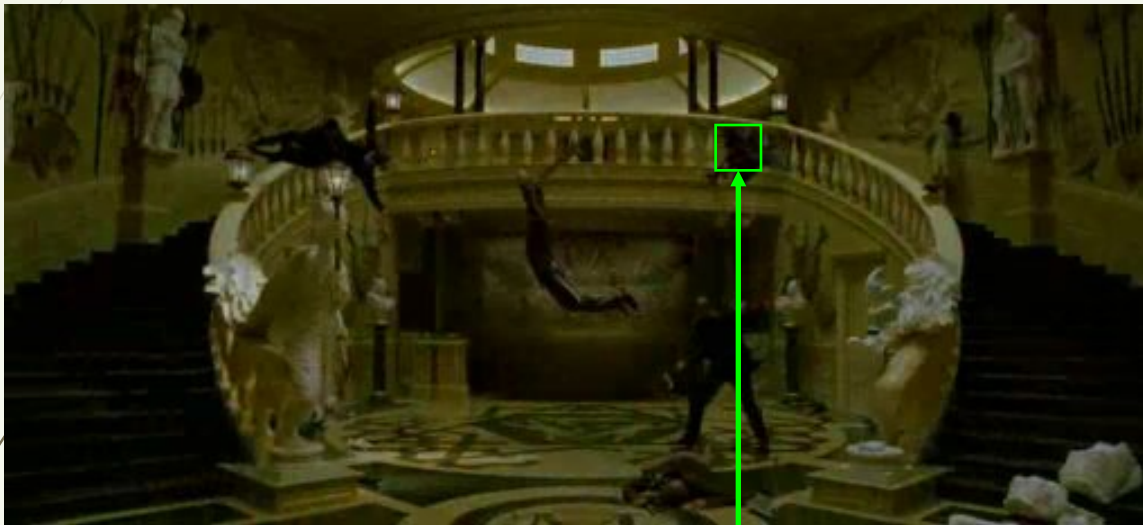
Essentially, jpeg is dct + quantization + entropy coding

- Essentially, JPEG is DCT + Quantization + Entropy Coding
- **Q1**: Which component is the lossy part? Quantization
- **Q2**: How do we control "how lossy is it"? Quantize ratio
- **Q3**: Can we use JPEG to encode video Yes as a sequence of images
No much redundancy is not leveraged
 - Yes, as a sequence of images
 - No, much (inter-frame) redundancy is not leveraged

Agenda

- Why Compression
- Image Coding Tools
- Video Coding Tools
- Popular Video Codecs

(Backward) Motion Estimation

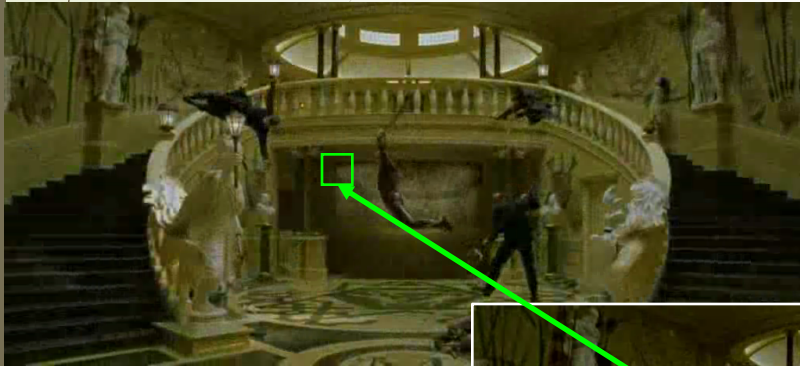


Frame t

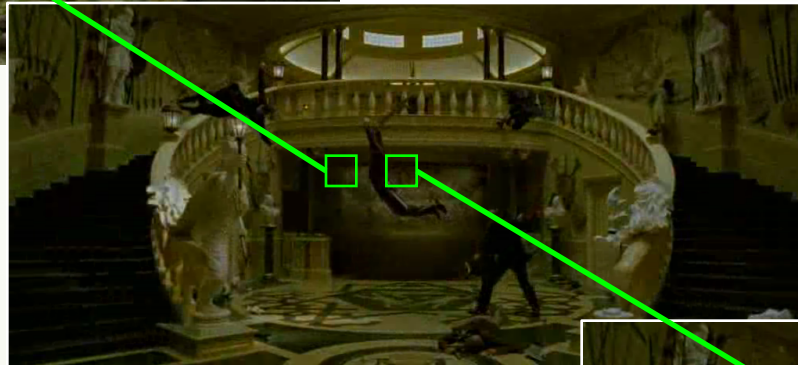


Frame $t+1$

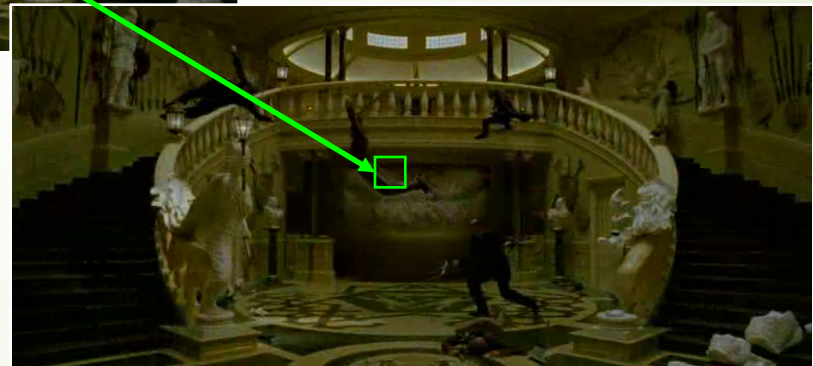
Bi-directional Prediction



Frame t

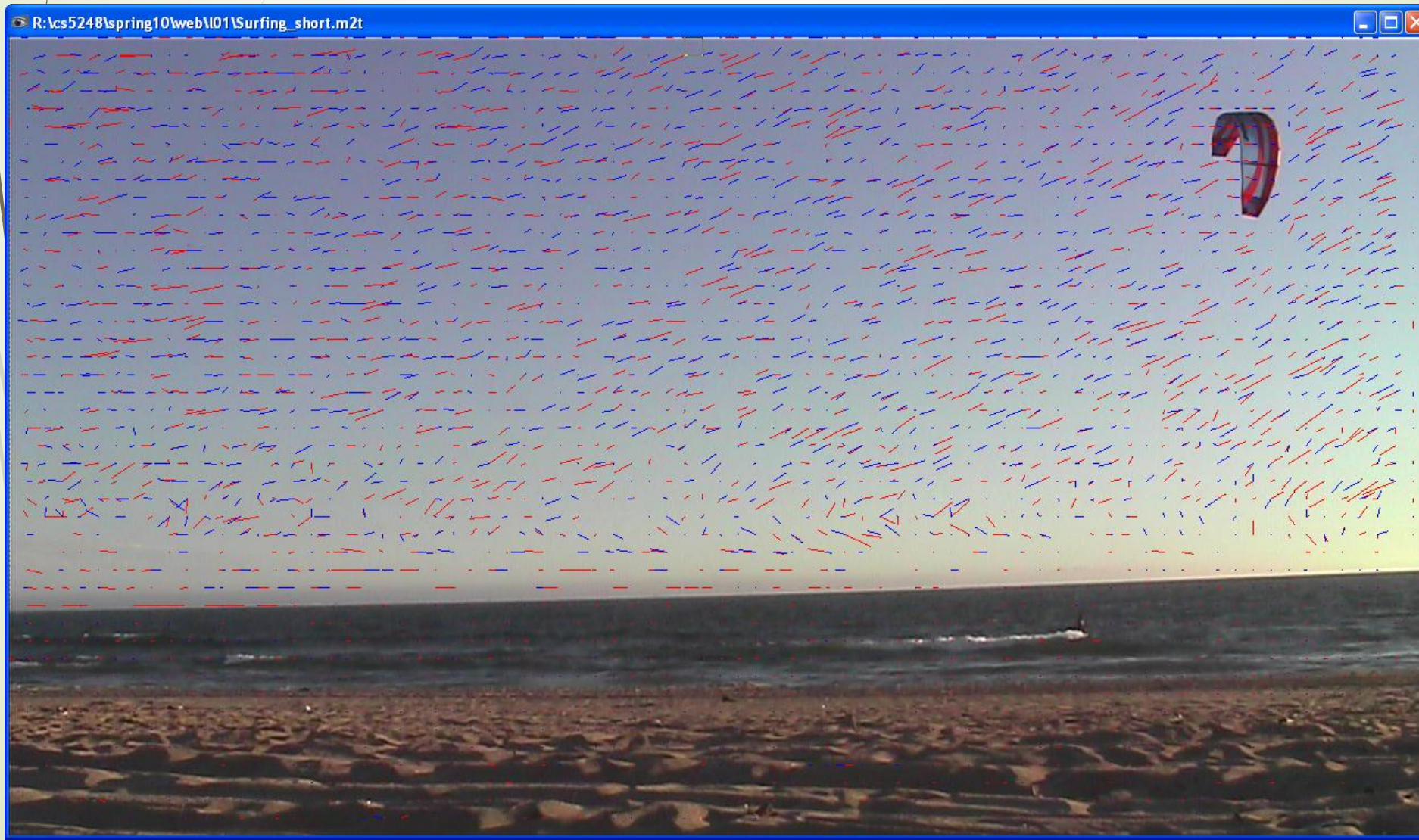


Frame $t+1$



Frame $t+2$

Sample Motion Vector



Original Frame 1



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Delta Between Frames 1 and 2



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Motion Compensated Difference

Idea: Just encode this “residue image”!

Such encoders (intra + inter-frame coding) are referred to as hybrid codecs